



Adaptive Differential Evolution-Based Distributed Model Predictive Control for Multi-UAV Formation Flight

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Abstract

The complicated formation flight for multi-unmanned aerial vehicles is a challenge, especially when multi-mission requirements are taken into account. This paper studies the adaptive differential evolution-based distributed model predictive control approach to deal with the multi-unmanned aerial vehicle flight achieving obstacle/collision avoidance and formation keeping simultaneously in the complex environment. Specifically, the distributed model predictive controller is designed to achieve stable flight for each unmanned aerial vehicle as well as taking the state and input saturation into account, where the local optimization problem is solved by the adaptive differential evolution algorithm. Besides, the adaptive adjustment to the prediction horizon for the model predictive controller is introduced, while the asymptotic convergence of the rolling optimization is analyzed as well. Finally, simulation examples are provided to illustrate the validity of the proposed control structure.

Keywords Distributed model predictive control (DMPC) · Adaptive differential evolution (DE) algorithm · Multi-unmanned aerial vehicle (UAV) system · Obstacle/collision avoidance · Formation keeping

1 Introduction

Over the past decades, the formation control issues of multi-unmanned aerial vehicle (UAV) systems have been extensively investigated, and tremendous achievements have been scored in both civilian and military areas, such as target search and localization [1], surveillance and reconnaissance [2], radiation detection and contour mapping [3] and so on. These flight missions not only impose stringent requirements over the modeling and control of UAV dynamics, but also increasingly require multiple UAVs to coordinate in a formation. Besides, in the complex flight environment, obstacle/collision avoidance is also necessary for UAV formation flight. Generally,

multiple UAV cooperation can be divided into centralized and distributed methods. In centralized approach, the global information is available for all UAVs, and there exists a control center assigning missions to each UAV. In [4], a centralized control method is proposed to command all UAVs achieving the accurate target localization. While in distributed method, by communicating with neighboring UAVs, each UAV could determine its own mission. In [5], a distributed control structure is adopted to finish target tracking for UAVs. To this end, plenty of methods for multi-UAV systems have been proposed recently, such as consensus control [6], model predictive control (MPC) [7], backstepping control [8], fault-tolerant control [9], optimal control [10], and robust control [11].

As mentioned above, MPC has been researched as one of the most popular advanced control schemes for UAV formation flight [12, 13]. It is known that the MPC strategy uses the designed model to predict the future behaviors and then an optimal control sequence can be obtained by an optimization of these predictions. This “foresee” feature of MPC makes it a potential control approach for UAV formation flight. Especially for the formation-keeping problem, MPC can take into account the future reference to improve the performance, such that both the current and future tracking errors can be suppressed simultaneously, as well as taking the state and input saturation into consideration.

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The performance of MPC highly relies on the local optimization problem, which can be solved by either analytical or numerical approach, where the cost function decides the solution. For the multi-UAV formation flight in this paper, the corresponding cost function is complex and non-differentiable. Traditional search techniques using the characteristics of the problem to determine the next sampling point (gradients, Hessians, and linearity) are no longer feasible, because the characteristics of flight are complex and constrained. Nevertheless, stochastic search techniques, such as genetic algorithm (GA), particle swarm optimization (PSO) algorithm and chaotic predator prey biogeography-based optimization (CPPBBO) algorithm [14–16], make no such solutions. Instead, their next point is determined by the stochastic decision rules. Therefore, this paper adopts a stochastic search technique based on the differential evolution (DE) algorithm [17] to solve the optimization problem in MPC, which has been successfully applied in many fields [18, 19].

Motivated by the above facts, this paper presents the adaptive DE-based distributed MPC strategy for the multi-UAV flight achieving obstacle/collision avoidance and formation keeping simultaneously. Although the application of DE-based MPC in multi-UAV system is not new [20, 21], we extend our work in the following contributions:

1. To obtain the better optimization results, adaptive adjustments are introduced for conventional DE algorithm.
2. Based on the complex flight environment, the optimizing index is designed as consensus tracking, obstacle avoidance, energy consumption and path smoothness, respectively. Besides, to improve the performance of obstacle avoidance, an adaptive adjustment to prediction horizon for MPC is considered.
3. A composite control structure combining MPC and adaptive DE is relatively novel for multi-UAVs to deal with obstacle/collision avoidance and formation keeping effectively, as well as taking the state and input saturation into consideration.

The remainder of this paper is organized as follows. The problem formulation and preliminaries are illustrated in Sect. 2. In Sect. 3, the adaptive DE-based MPC strategy is elaborated. Asymptotic convergence analysis of the rolling optimization is illustrated in Sect. 4. Section 5 carries out the simulation studies. Finally, conclusions and future works are presented in Sect. 6.

2 Problem Statement and Preliminaries

2.1 Kinematic and Dynamic Models

Suppose that there are N ($N > 1$) UAVs with the same dynamic characteristics moving in $\mathbb{R}^3$, which together compose a multi-UAV system. In what follows, the dynamic model supposes that the UAV always performs coordinated maneuvers and its thrust is directed along the velocity vector. Also, assume that the earth is flat and the mass center is time invariant. Under these assumptions, the motion equations of the UAV can be described as follows [22]:

$$\begin{aligned}\dot{x}_i &= v_i \cos \gamma_i \cos \psi_i, \\ \dot{y}_i &= v_i \cos \gamma_i \sin \psi_i, \\ \dot{z}_i &= v_i \sin \gamma_i,\end{aligned}\quad (1)$$

where $i = 1, \dots, N$ is the index of UAVs, (x_i, y_i, z_i) denotes the position coordinates, v_i is the velocity, and γ_i and ψ_i are the flight path angle and heading angle, respectively. Then the corresponding dynamics are presented by

$$\begin{aligned}\dot{v}_i &= (T_i - D_i)/m_i - g \sin \gamma_i, \\ \dot{\psi}_i &= \frac{L_i \sin \phi_i}{m_i v_i \cos \gamma_i}, \\ \dot{\gamma}_i &= \frac{g}{v_i} (n_i \cos \phi_i - \cos \gamma_i),\end{aligned}\quad (2)$$

where m_i is the mass, g is the gravitational acceleration, D_i is the drag force, L_i is the lift force, T_i is the engine thrust controlled by the throttle, ϕ_i is the bank angle controlled by a combination of rudder and ailerons, and $n_i = L_i/m_i g$ is the load factor controlled by the elevator. During the flight, all the control variables should be constrained within their respective limits.

Based on that, define $a_{ti} = (T_i - D_i)/m_i$, $a_{yi} = g n_i \sin \phi_i$ and $a_{pi} = g n_i \cos \phi_i$. Then the control input is presented as $\mathbf{u}_i = [a_{ti}, a_{yi}, a_{pi}]^T$, and the state variable is defined by $\mathbf{x}_i = [v_i, \psi_i, \gamma_i, x_i, y_i, z_i]^T$. Therefore, the kinematics of the i th UAV can be described as follows:

$$\dot{\mathbf{x}}_i(t) = \mathbf{f}(\mathbf{x}_i(t), \mathbf{u}_i(t)), \quad i = 1, \dots, N. \quad (3)$$

On the basis of (3), if the control input $\mathbf{u}_i$ and the initial state $\mathbf{x}_0$ are provided, the UAV model during each time horizon $t \in (0, T]$ is obtained as follows:

$$\begin{cases} \mathbf{x}_i(t) = \mathbf{x}_0 + \int_0^t \mathbf{f}(\mathbf{x}_i(\tau), \mathbf{u}_i(\tau)) d\tau \\ \mathbf{y}_i(t) = \mathbf{g}(\mathbf{x}_i(t), t) \end{cases}. \quad (4)$$

2.2 Formation Structure Definition

In this paper, the leader–follower structure is proposed in Fig. 1. Each UAV is equipped with a complete set of communication equipment and controller. Set one UAV as the leader of the formation, which is treated as the reference point, and its state vector is presented by $\mathbf{x}_L$. Assuming that the information exchange topology is undirected and connected, each follower exchanges the state information with the leader directly or indirectly, and executes the corresponding maneuvers to keep the relative state $(\mathbf{x}_i - \mathbf{x}_L)$ to the leader.

The distance between the i th UAV and r th UAV can be described as

$$d_{ir}(t) = \sqrt{(x_i(t) - x_r(t))^2 + (y_i(t) - y_r(t))^2 + (z_i(t) - z_r(t))^2}. \quad (5)$$

To avoid collisions among UAVs, $d_{ir}(t)$ should be longer than the safe anti-collision distance D_{safe} , i.e.,

$$d_{ir}(t) \geq D_{\text{safe}}, \quad \forall i, r \in \{1, \dots, N\}. \quad (6)$$

Similarly, to ensure the real-time communication among the adjacent UAVs, $d_{ir}(t)$ should be shorter than the communication distance D_{com} , i.e.,

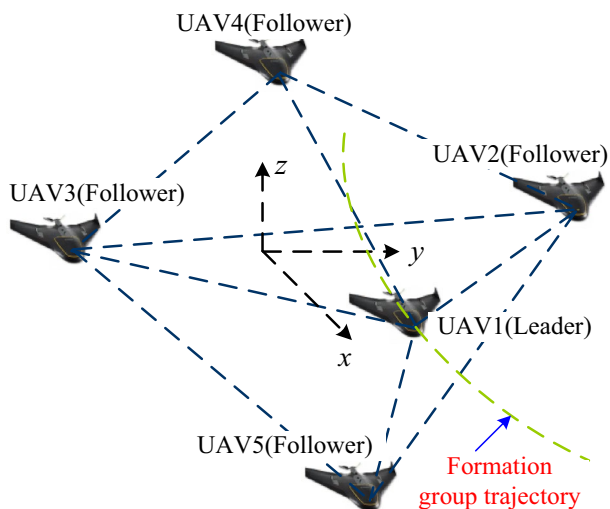
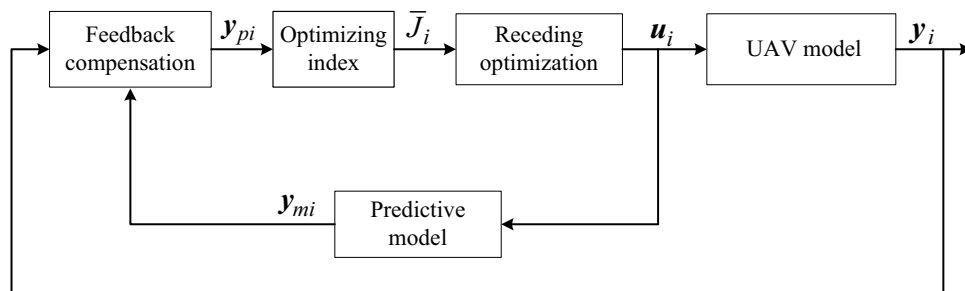


Fig. 1 Multi-UAV in leader–follower structure

Fig. 2 Control structure of MPC



$$d_{ir}(t) \leq D_{\text{com}}, \quad \exists i, r \in \{1, \dots, N\}. \quad (7)$$

3 Adaptive DE-Based MPC for Formation Flight

Generally, MPC method is a p -step-ahead online optimization strategy, and its corresponding structure is shown in Fig. 2. At each time step, MPC method optimizes the control problem for the following p steps, and only the first element of the optimal control sequence is implemented to the predictive model, where the control horizon is m . Then the predictive output $\mathbf{y}_{mi}$ is obtained. Through feedback compensation, the revised predictive output $\mathbf{y}_{pi}$ is determined. The control input $\mathbf{u}_i$ is generated by the receding optimization of the optimizing index $\bar{J}_i$. Then the system output $\mathbf{y}_i$ is obtained. The whole process will be iterated online to achieve distributed control for each UAV [23].

3.1 Predictive Model

Define T as the sampling time. Using the Euler's method, system (4) can be recast into a discrete model as follows:

$$\begin{cases} \mathbf{x}_i(k+1) = \mathbf{x}_i(k) + \int_{kT}^{(k+1)T} \mathbf{f}(\mathbf{x}_i(t), \mathbf{u}_i(k)) dt \\ \mathbf{y}_{mi}(k) = \mathbf{C}\mathbf{x}_i(k) \end{cases}, \quad (8)$$

subject to

$$\begin{aligned} \mathbf{u}_{\min} &\leq \mathbf{u}_i(k) \leq \mathbf{u}_{\max}, \\ \mathbf{x}_{\min} &\leq \mathbf{x}_i(k) \leq \mathbf{x}_{\max}, \end{aligned} \quad (9)$$

where $\mathbf{x}_i(k) = [v_i(k), \psi_i(k), \gamma_i(k), x_i(k), y_i(k), z_i(k)]^T$ is the initial state at the k th time step, $\mathbf{u}_i(k) = [a_{ii}(k), a_{yi}(k), a_{pi}(k)]^T$ is the control input, which stays constant during each time interval $[kT, (k+1)T]$, and $\mathbf{C} = [\mathbf{0}_{3 \times 3} \quad \mathbf{I}_{3 \times 3}]$.

3.2 Feedback Compensation

Based on the p -step-ahead online optimization of MPC, the predictive control sequence is computed by $\mathbf{u}_i(k|k)$ and $\mathbf{u}_i(k+1|k), \dots, \mathbf{u}_i(k+p-1|k)$. The predictive outputs are

obtained by $y_{mi}(k+1|k)$, $y_{mi}(k+2|k)$, ..., $y_{mi}(k+p|k)$. Due to the possible external disturbances, the predictive output $y_{mi}(k+j|k)$ is hard to match the actual output $y_i(k+j)$. Therefore, the predictive output should be further revised as follows:

$$y_{pi}(k+j|k) = y_{mi}(k+j|k) + H e_i(k), \quad (10)$$

where $H = I_{3 \times 3}$ is the error coefficient matrix, and $e_i(k)$ is defined as

$$e_i(k) = y_i(k) - y_{mi}(k|k-1). \quad (11)$$

3.3 The Optimizing Index

Considering the formation flight for each UAV, the optimizing index is introduced by

$$\min J_i(k) = \lambda_1 J_i^1(k) + \lambda_2 J_i^2(k) + \lambda_3 J_i^3(k) + \lambda_4 J_i^4(k), \quad (12)$$

where $J_i^1(k)$, $J_i^2(k)$, $J_i^3(k)$ and $J_i^4(k)$ represent the consensus tracking cost, obstacle avoidance cost, control effort cost and path smoothness cost, respectively. λ_1 , λ_2 , λ_3 , and λ_4 are weighting factors, satisfying $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$.

$J_i^1(k)$ has the form of

$$J_i^1(k) = \sum_{j=1}^p \|y_{pi}(k+j|k) - r_i(k+j)\|_{\mathcal{Q}}^2, \quad (13)$$

subject to

$$y_{\min} \leq y_{pi}(k+j|k) \leq y_{\max}, \quad (14)$$

where p is the prediction horizon, $\|\cdot\|_{\mathcal{Q}}^2$ are the squared 2-norms weighted by positive definite matrices $\mathcal{Q}$.

Remark 1 If the i th UAV is the leader, $r_i(k)$ denotes the given trajectory. On the other hand, if the i th UAV is the follower, $r_i(k)$ represents the relative position information to the leader.

$J_i^2(k)$ is defined by

$$J_i^2(k) = \sum_{l=1}^O \sum_{j=1}^p \chi_l(k+j|k),$$

$$\chi_l(k+j|k) = \begin{cases} \frac{1}{\|y_{pi}(k+j|k) - \eta_l\|_G^2 + \varepsilon}, & \|y_{pi}(k+j|k) - \eta_l\| < R_{\text{warning}} \\ 0, & \|y_{pi}(k+j|k) - \eta_l\| \geq R_{\text{warning}} \end{cases}, \quad (15)$$

where O is the number of obstacles in the detection region, $\eta_l = (x_l, y_l, z_l)$ is the central position of the l th obstacle, G is a positive definite matrix which is adjusted based on $\|y_{pi}(k+j|k) - \eta_l\|$ in warning and dangerous region, given by G_1 and G_2 , respectively. ε is a positive quantity to prevent $J_i^2(k)$ from being singular when $\|y_{pi}(k+j|k) - \eta_l\| \rightarrow 0$.

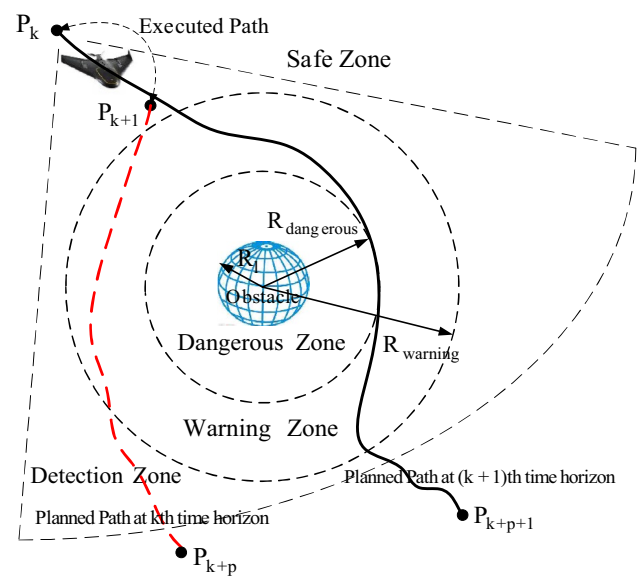


Fig. 3 Collision avoidance for the UAV

Remark 2 As shown in Fig. 3, the behavior of the collision avoidance relates to the measured distance to the detected obstacles. Specifically, when the UAV moves in the warning zone, the autonomous obstacle avoidance is activated. According to the measured distance to the obstacles, the control action of each UAV is computed within each time horizon. Especially close to the dangerous zone, the control force against obstacles becomes stronger.

$J_i^3(k)$ has the regular quadratic form of

$$J_i^3(k) = \sum_{j=1}^m \|u_i(k+j-1|k)\|_R^2, \quad (16)$$

subject to

$$u_{\min} \leq u_i(k+j-1|k) \leq u_{\max}, \quad (17)$$

where m ($1 \leq m \leq p$) is the control horizon.

$J_i^4(k)$ is introduced as follows [24]:

$$J_i^4(k) = \frac{1}{p} \sum_{j=1}^p (|\psi_i(k+j|k) - \psi_i(k+j-1|k)| + |\gamma_i(k+j|k) - \gamma_i(k+j-1|k)|), \quad (18)$$

subject to

$$\begin{aligned} \psi_{\min} &\leq \psi_i(k+j|k) \leq \psi_{\max}, \\ \gamma_{\min} &\leq \gamma_i(k+j|k) \leq \gamma_{\max}. \end{aligned} \quad (19)$$

Remark 3 $J_i^4(k)$ denotes the average change of path angle, reflecting the smoothness of flight path. And this introduction penalizes large maneuvers during the obstacle avoidance.

Considering the inequality constraints in (14), (17) and (19), a feasible approach for constraints is to transform the original problem to its alternative unconstrained form using a penalty cost. Then we substitute $l_1(k+j|k) = y_{\max} - y_{pi}(k+j|k)$, $l_2(k+j|k) = y_{pi}(k+j|k) - y_{\min}$, $h_1(k+j|k) = x_{\max} - x_i(k+j|k)$, $h_2(k+j|k) = x_i(k+j|k) - x_{\min}$, $m_1(k+j-1|k) = u_{\max} - u_i(k+j-1|k)$ and $m_2(k+j-1|k) = u_i(k+j-1|k) - u_{\min}$. Based on that, the penalty cost function can be described as [25]

$$\begin{aligned} \bar{J}_i(k) = & J_i(k) + \alpha \sum_{j=1}^{H_p} \|l_1(k+j|k)\|_{S(l_1(k+j|k))}^2 \\ & + \alpha \sum_{j=1}^{H_p} \|l_2(k+j|k)\|_{S(l_2(k+j|k))}^2 \\ & + \beta \sum_{j=1}^{H_u} \|h_1(k+j|k)\|_{S(h_1(k+j|k))}^2 \\ & + \beta \sum_{j=1}^{H_u} \|h_2(k+j|k)\|_{S(h_2(k+j|k))}^2 \\ & + \eta \sum_{j=1}^{H_u} \|m_1(k+j-1|k)\|_{S(m_1(k+j-1|k))}^2 \\ & + \eta \sum_{j=1}^{H_u} \|m_2(k+j-1|k)\|_{S(m_2(k+j-1|k))}^2, \end{aligned} \quad (20)$$

where $S(a) = \begin{bmatrix} s_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & s_n \end{bmatrix}$, with $a = [a_1, \dots, a_n]^T$, and if

$a_i < 0$, $s_i = 10^5$; otherwise, $s_i = 0$, $i = 1, \dots, n$. The introduction of $S(a)$ makes it possible to consider the inequality constraints during the optimization.

In addition, based on (6) and (7), the constraints of D_{safe} and D_{com} are also considered. The corresponding auxiliary optimization term is

$$\begin{aligned} P_i(k) = & \kappa \cdot \sum_{j=1}^p \sum_{i \neq r} [\max(0, D_{\text{safe}} - d_{ir}(k+j|k)) \\ & + \max(0, d_{ir}(k+j|k) - D_{\text{com}})]^2, \end{aligned} \quad (21)$$

where $\kappa = 10^5$. According to the relative position of multiple UAVs, $P_i(k)$ is added in (20).

3.4 Receding Optimization

Since the optimizing index $\bar{J}_i(k)$ is complex and non-differentiable, using the traditional search techniques is difficult to solve the optimization problem. Referring to [26], the population-based optimization approaches provide good solutions to similar complicated problems. Inspired by the

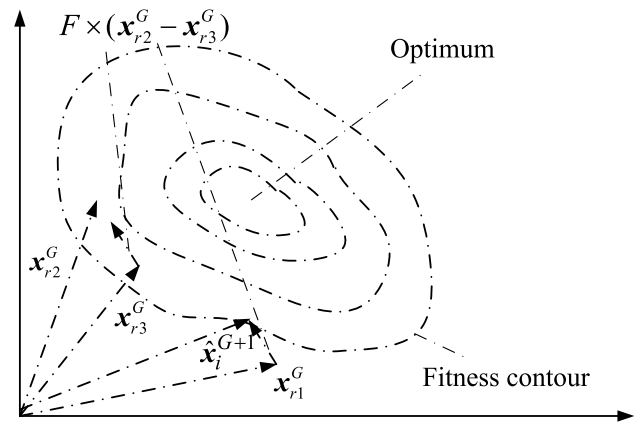


Fig. 4 Mutation process of DE algorithm

fact, differential evolution (DE) algorithm is utilized to deal with the receding optimization for MPC. The goal is to find the proper control variable that minimizes the optimizing index $\bar{J}_i(k)$.

DE algorithm is an optimization approach which searches randomly in continuous space with three evolutionary operations in the following, namely mutation, recombination and selection [17]. Suppose that the initial population has N_p individuals inside the given D -dimensional search space. The individual vector in continuous space can be presented by $x_i^G = (x_{i1}, x_{i2}, \dots, x_{iD})$, $i = 1, \dots, N_p$, and $G = 1, \dots, N_G$.

3.4.1 Mutation Operation

As shown in Fig. 4, for each x_i^G in the parent population, the mutation individual $\hat{x}_i^{G+1}$ is produced by

$$\hat{x}_i^{G+1} = x_{r1}^G + F \times (x_{r2}^G - x_{r3}^G), \quad (22)$$

where r_1, r_2 and r_3 are randomly chosen and different from the running index i , and F is a constant scaling factor within $[0, 1]$, which controls the amplification of the differential variation $(x_{r2}^G - x_{r3}^G)$.

3.4.2 Recombination Operation

For each x_i^G , a trial individual $\bar{x}_i^{G+1}$ is generated by coping components from the mutation vector $\hat{x}_i^{G+1}$ and target vector x_i^G in dependence. As shown in Fig. 5, it can be formulated as follows:

$$\bar{x}_{ij}^{G+1} = \begin{cases} \hat{x}_{ij}^{G+1}, & \text{if } r_b \leq C_R, \quad i = 1, \dots, N_p \\ x_{ij}^G, & \text{otherwise,} \quad j = 1, \dots, D \end{cases}, \quad (23)$$

where $r_b \in [0, 1]$ is a uniform random number, and C_R is a constant within $[0, 1]$.

3.4.3 Selection Operation

Offspring $\bar{x}_i^{G+1}$ competes one-to-one with its parent x_i^G . The selection operation is expressed as

$$x_i^{G+1} = \begin{cases} \bar{x}_i^{G+1}, & \text{if } f(\bar{x}_i^{G+1}) \leq f(x_i^G), \\ x_i^G, & \text{otherwise} \end{cases} \quad (24)$$

This selection scheme ensures that the best individual with the minimum cost can be retained when moving from one generation to the next, leading to a fast convergence behavior.

In spite of the effectiveness of DE algorithm in wide ranges [27], conventional algorithm alone cannot deal with the UAV flight variation effectively, since selecting suitable mutation strategies and control parameters at different evolution stages is difficult. Therefore, in this paper, a self-adaptive DE algorithm is proposed, where adaptive improvements for F and C_R are introduced as follows:

1. Adaptive improvement for F

$$F_m = \begin{cases} F_1, & F_0 \times \sum_{i=1}^{N_p} \bar{J}(x_i^m) / \sum_{i=1}^{N_p} \bar{J}(x_i^0) \leq F_1 \\ F_0 \times \frac{\sum_{i=1}^{N_p} \bar{J}(x_i^m)}{\sum_{i=1}^{N_p} \bar{J}(x_i^0)}, & F_0 \times \sum_{i=1}^{N_p} \bar{J}(x_i^m) / \sum_{i=1}^{N_p} \bar{J}(x_i^0) > F_1 \end{cases}, \quad (25)$$

where F_0 and F_m represent the F of the initial iteration and the m th iteration, respectively. $\sum_{i=1}^{N_p} \bar{J}(x_i^0)$ and $\sum_{i=1}^{N_p} \bar{J}(x_i^m)$ denote the costs of the initial iteration and the m th iteration, respectively.

2. Adaptive improvement for C_R

$$C_{R_n} = \begin{cases} C_{R_1}, & \bar{J}(x_n^G) > \bar{J}(x_{\text{avg}}^G) \\ C_{R_0} \times \frac{(C_{R_1} - C_{R_0})(\bar{J}(x_{\text{avg}}^G) - \bar{J}(x_n^G))}{\bar{J}(x_{\text{avg}}^G) - \bar{J}(x_{\text{min}}^G)}, & \bar{J}(x_n^G) \leq \bar{J}(x_{\text{avg}}^G) \end{cases}, \quad (26)$$

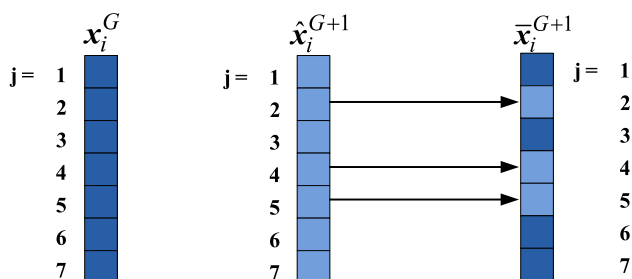


Fig. 5 Recombination process of DE algorithm

where $\bar{J}(x_n^G)$ is the n th individual cost, $\bar{J}(x_{\text{avg}}^G) = \sum_{i=1}^{N_p} \bar{J}(x_i^G) / N_p$ is the average cost of all individuals, and $\bar{J}(x_{\text{min}}^G) = \min\{\bar{J}(x_1^G), \dots, \bar{J}(x_{N_p}^G)\}$ is the minimum cost of all individuals.

The control algorithm is summarized as follows:

- Step 1. Load the initial state of the UAV and its corresponding reference. Then initialize the population, where each individual in population is a candidate solution to $U_i(k)$, i.e., $U_i(k) = [u_i(k|k), u_i(k+1|k), \dots, u_i(k+p-1|k)]$.
- Step 2. Each individual carries out the predictive model and feedback compensation to obtain the predictive output $y_{pi}(k+j|k)$.
- Step 3. Using the predictive output $y_{pi}(k+j|k)$, calculate the cost for each individual. Then the individual with the minimum cost is recorded.
- Step 4. Execute mutation, recombination to create new individuals for the next generation.
- Step 5. Obtain the prediction output of each new individual and calculate their respective costs.
- Step 6. Perform the selection operation for each individual to ensure the better individual can remain in the population.
- Step 7. Repeat Steps 3–6 until the convergence criteria of the control algorithm is met.
- Step 8. The best individual $U_i^*(k)$ is determined, and its first element $u_i^*(k|k)$ is implemented to both the predictive model and UAV system. The corresponding outputs are obtained.
- Step 9. Go to Step 1 for the next time horizon.

3.5 Adaptive Adjustments to Prediction Horizon

The prediction horizon p is the crux of MPC method, which decides the optimization performance to a large extent. If p is small, the performance will be of poor optimization since the future motions cannot be fully considered. As p increases, the prediction performance would improve. But if p is too large, the computational cost would increase significantly. Therefore, a proper selection of p is important, especially for multi-UAVs in obstacle-laden environment. According to the number of obstacles in the detection region, we define

$$p = O + 1. \quad (27)$$

Remark 4 If there are few obstacles in the detection area, p will be reduced to save the unnecessary computational cost since the UAV can easily achieve obstacle avoidance. On the contrary, if there are numerous obstacles in the detection region, it is a big challenge for safe flight of the UAV. As a result, p will increase to take the multi-step motional tendency into account.

4 Convergence Analysis

It is known that the MPC method divides the control problem into a series of local optimizations, and each local optimization is solved by the adaptive DE algorithm. In this section, the asymptotic convergence of the local optimization is first analyzed; on this basis, the global convergence analysis is presented as well.

4.1 Local Convergence Analysis [28]

Suppose that at the k th time horizon, the optimal solution set of the cost function $\bar{J}(k)$ can be defined as follows:

$$S^*(k) = \{\mathbf{x}^* : \bar{J}(\mathbf{x}^*|k) = \min\{\bar{J}(\mathbf{x}|k) : \mathbf{x} \in S(k)\}\}, \quad (28)$$

where $S(k)$ is a measurable search space and $\mathbf{x}^*$ represents the optimal solution individual.

However, condition (28) is hard to be satisfied in actual process. Thus, an expanded optimal solution set is considered as follows:

$$S_\varepsilon^*(k) = \{\tilde{\mathbf{x}} \in S(k) : |\bar{J}(\tilde{\mathbf{x}}|k) - \bar{J}(\mathbf{x}^*|k)| < \varepsilon\}, \quad (29)$$

where $\varepsilon > 0$ is a small positive real number. Denoting μ as the Lebesgue measure, we suppose that the $\mu(S_\varepsilon^*(k)) > 0$ for each ε . Expanded optimal solution (29) can be approximately regarded as the optimal one.

Before analyzing the convergence of local optimization, we need to introduce a concept of the convergence in probability as follows:

Define $\{X_G, G = 1, 2, \dots\}$ as a population sequence generated by the adaptive DE algorithm for the local optimization problem, where $X_G = \{\mathbf{x}_i^G, i = 1, 2, \dots, N_p\}$. It is required that the algorithm converges to the optimal solution set in probability if and only if

$$\lim_{G \rightarrow \infty} P\{X_G \cap S_\varepsilon^*(k) \neq \emptyset\} = 1, \quad (30)$$

where P denotes the probability of an event.

Assumption 1 Based on the above definition, it is assumed that for each population X_G of the adaptive DE algorithm, there exists at least one individual $\mathbf{x}^G$ such that

$$P\{\mathbf{x}^G \in S_\varepsilon^*(k)\} \geq \alpha > 0, \quad (31)$$

where α is a small positive value.

Proof 1 Suppose that an individual $\mathbf{x}_{\text{rand}}^G$ is generated randomly in each population X_G . The probability that $\mathbf{x}_{\text{rand}}^G$ belongs to the optimal solution set is given by

$$P\{\mathbf{x}_{\text{rand}}^G \in S_\varepsilon^*(k)\} = \frac{\mu(S_\varepsilon^*(k))}{\mu(S(k))} = \alpha > 0. \quad (32)$$

It means that $P\{\mathbf{x}_{\text{rand}}^G \notin S_\varepsilon^*(k)\} = 1 - \alpha > 0$. The probability that the first d populations do not include an individual $\mathbf{x}^G \in S_\varepsilon^*(k)$ can be estimated by

$$\prod_{G=1}^d P\{X_G \cap S_\varepsilon^*(k) = \emptyset\} \leq (1 - \alpha)^d. \quad (33)$$

Based on basic ideas of the DE algorithm, the best individual in the population X_G has the same or better fitness value than the best one from all the previous populations, implying

$$\begin{aligned} \lim_{G \rightarrow \infty} P\{X_G \cap S_\varepsilon^*(k) = \emptyset\} \\ = \lim_{d \rightarrow \infty} \prod_{G=1}^d P\{X_G \cap S_\varepsilon^*(k) = \emptyset\} \leq \lim_{d \rightarrow \infty} (1 - \alpha)^d = 0, \end{aligned} \quad (34)$$

in other words,

$$\begin{aligned} \lim_{G \rightarrow \infty} P\{X_G \cap S_\varepsilon^*(k) \neq \emptyset\} \\ = 1 - \lim_{d \rightarrow \infty} \prod_{G=1}^d P\{X_G \cap S_\varepsilon^*(k) = \emptyset\} = 1 - 0 = 1, \end{aligned} \quad (35)$$

which proves (40).

4.2 Global Convergence Analysis

Based on the basic ideas of the MPC approach, the local optimizations follow the same control algorithm using the adaptive DE algorithm. Therefore, the local convergence analysis of the k th time horizon can be applicable to the $(k + j)$ th time horizon, $j = 1, 2, \dots$. Furthermore, the local optimization of the $(k + j)$ th time horizon is executed on the basis of the optimization results of the $(k + j - 1)$ th time horizon. Finally, this local convergence of each time horizon leads to the global convergence of the entire control process.

5 Simulation Analysis

In this section, the simulations are performed to verify the effectiveness of the proposed control strategy. The drag for each UAV in this paper is given by [29]

$$D_i = \frac{0.5\rho(v_i - v_{iw})^2 A_{\text{rea}} C_{D0} + 2k_d k_n^2 n_i^2 m_i^2}{[\rho(v_i - v_{iw})^2 A_{\text{rea}}]}, \quad (36)$$

where the dynamic parameters of the UAV model are shown in Table 1. The gust model $v_{iw} = v_{iw, \text{normal}} + v_{iw, \text{tan}}$ is obtained and varies according to the altitude z_i , and the normal wind shear is given by

Table 1 Dynamic parameters

Parameters	Value
Mass m_i (kg)	20
Wing area A_{rea} (m^2)	1.6
Zero-lift drag coefficient C_{D0}	0.02
Load factor effectiveness k_n	1
Induced drag coefficient k_d	1.25
Atmospheric density ρ (kg/m^3)	1.225
Gravitational coefficient g (kg/m^2)	9.81

$$v_{iw,normal} = 0.215v_m \lg(z_i) + 0.285v_m, \quad (37)$$

where $v_m = 4m/s$ is the average wind speed at the altitude of 80 m. The turbulence part of the wind gust $v_{iw,tan}$ has a Gaussian distribution with a zero mean and a standard derivation of $0.09v_m$.

The constraints of the system state z_i , v_i , ψ_i , γ_i , and control variables a_{ii} , a_{yi} , a_{pi} are shown in Table 2. Besides, the relevant parameters are set as follows: $F_0 = 0.6$, $F_1 = 0.2$, $C_{R_0} = 0.6$, $C_{R_1} = 0.9$, $N_p = N_c = 50$, $\lambda_1 = \lambda_2 = 0.4$, $\lambda_3 = \lambda_4 = 0.1$, $\alpha = \beta = \eta = 1$, $Q = I_{3 \times 3}$, $R = I_{3 \times 3}$, $G_1 = 0.005I_{3 \times 3}$, $G_2 = 0.0005I_{3 \times 3}$ and $T = 0.5s$.

To test the proposed control scheme, a flight scenario is constructed for a multi-UAV formation. Assuming that a triangular formation is composed of five UAVs, the relative positions of each UAV are described in Fig. 6. All five UAVs are initially flying at the speed of 50 m/s and at the altitude of 100 m. The initial positions of five UAVs are $[-20, 20, 100]$ m, $[-10, 10, 100]$ m, $[0, 0, 100]$ m, $[-10, -10, 100]$ m, and $[-20, -20, 100]$ m, respectively. Suppose the third UAV as the leader tracks a reference trajectory: $x = 50 \cdot t$, $y = 0$, $z = 100$, $t \in [0, 5]$. And other UAVs as the followers fly with the relative position: $\Delta y_{1,3} = [-20, 20, 0]$ m, $\Delta y_{2,3} = [-10, 10, 0]$ m, $\Delta y_{3,4} = [-10, -10, 0]$ m, and $\Delta y_{3,5} = [-20, -20, 0]$ m. Besides, the anti-collision radius, communication radius and detection radius of UAV are assumed to be 5 m, 50 m,

and 150 m, respectively. Additionally, an obstacle-laden environment is described, where each obstacle is assumed to be a sphere.

$$\xi = \bigcup_{l=1}^{l=4} \xi_l, \quad (38)$$

where ξ_l denotes the obstacle space, which is a function about the radius R_l and the central position (x_l, y_l, z_l) of the obstacle, i.e., $\xi_l = f((x_l, y_l, z_l), R_l)$. In this section, define $\xi_1 = f((80, 0, 100)m, 15m)$, $\xi_2 = f((180, 0, 125)m, 22m)$, $\xi_3 = f((230, 40, 100)m, 20m)$, and $\xi_4 = f((120, -60, -120)m, 30m)$. Besides, suppose $R_{dangerous} = R_l + 50m$ and $R_{warning} = R_l + 100m$.

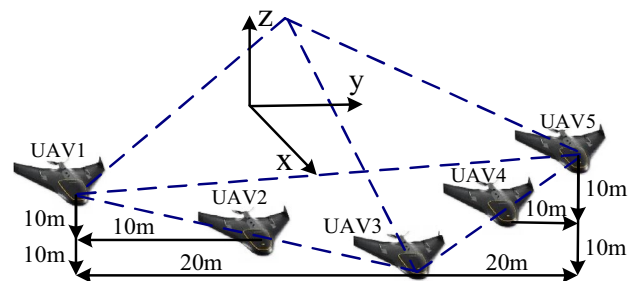
The 3D performances of the UAV formation flight with the proposed control method are depicted graphically in Figs. 7 and 8. Generally, the proposed control scheme manages to achieve formation keeping and obstacle avoidance simultaneously. However, for $p = 1$ in Fig. 7, some large angle changes and formation keeping errors exist in the flight performance, even the formation consensus cannot be achieved well after avoiding obstacles within the prescribed time. On the contrary, in Fig. 8, the flight performance of adaptive p is better with smoother path planning and less formation keeping errors. The corresponding numerical analyses are shown in Tables 3 and 4. Besides, the control actions are presented in Fig. 9, and the adaptive selection of p is shown in Fig. 10.

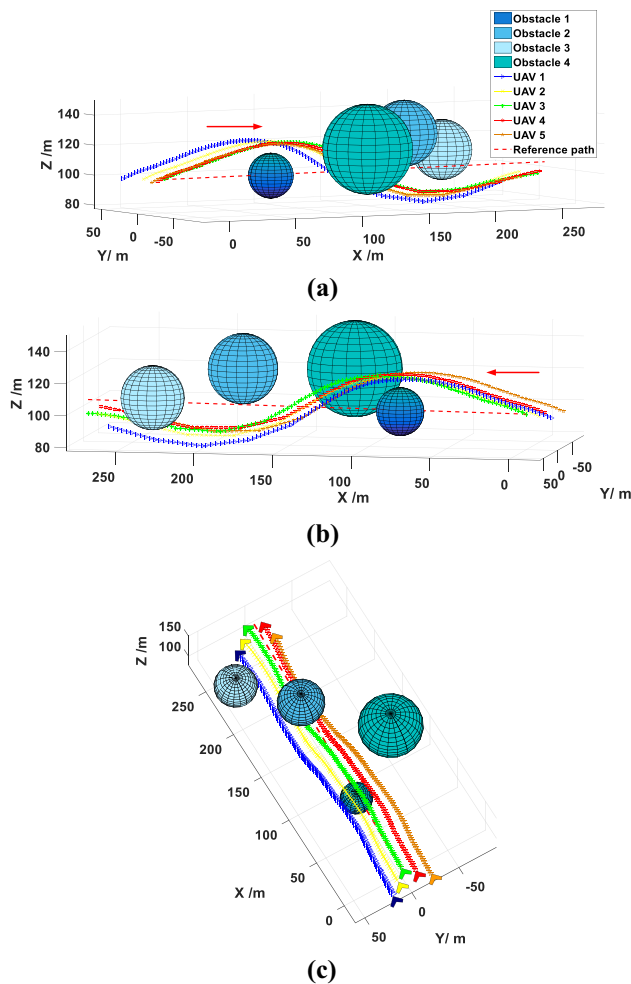
To further analyze the tracking performances, the integral of absolute error (IAE) is introduced in (39), and the comparative results are shown in Table 3. It can be seen that the adaptive p strategy provides a more satisfied formation tracking performance:

$$IAE = \frac{1}{N \cdot H_T} \sum_{i=1}^N \sum_{k=1}^{H_T} \|y_i(k) - r_i(k)\|_{I_{3 \times 3}}^2, \quad (39)$$

Table 2 Constraints of system states and control variables

Parameters	Value
Altitude z_i (m)	[50, 1000]
Velocity v_i (m/s)	[0, 100]
Heading angle ψ_i ($^\circ$)	$[-180, 180]$
Flight path angle γ_i ($^\circ$)	$[-30, 30]$
Control variable a_{ii} (m/s^2)	$[-2, 4]$
Control variable a_{yi} (rad/s^2)	$[-15, 15]$
Control variable a_{pi} (rad/s^2)	$[-11, 15]$

**Fig. 6** Five UAVs in formation flight

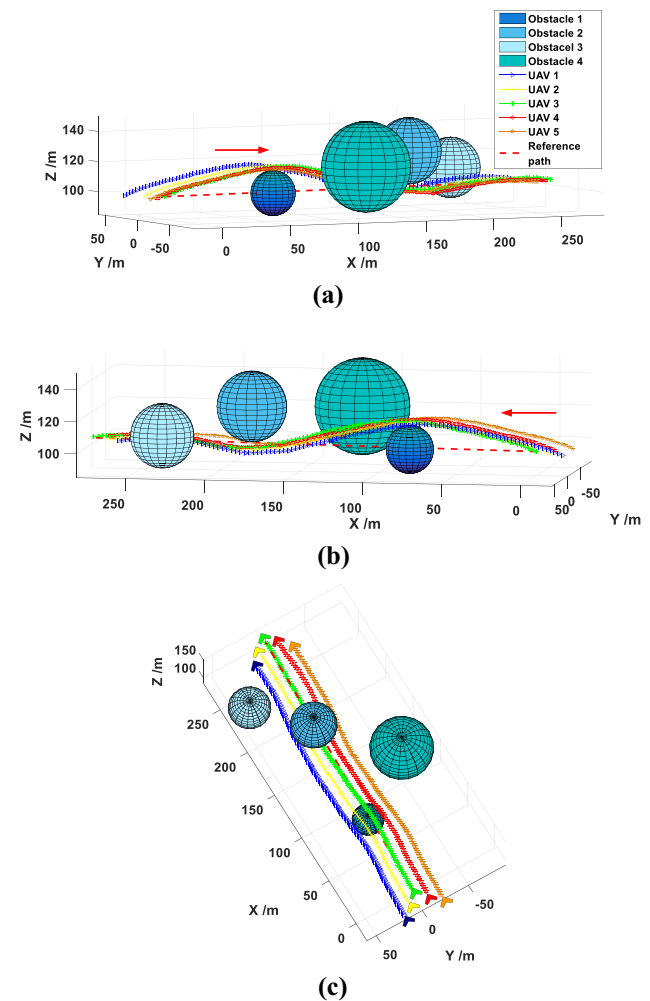
Fig. 7 3D performance of $p = 1$

where N is the total number of UAVs and H_T is the total number of time horizons.

Besides, to evaluate the performance of path planning, the global smoothness (GS) is defined in (40). The corresponding results are shown in Table 4. From the table, it can be judged that the adaptive p strategy provides a smoother flight path.

$$GS = \frac{1}{N \cdot H_T} \cdot \sum_{i=1}^N \sum_{k=1}^{H_T} (|\psi_i(k+1) - \psi_i(k)| + |\gamma_i(k+1) - \gamma_i(k)|). \quad (40)$$

The proposed MPC algorithm is solved in computer with Intel(R) Core(TM) i5-9400 CPU @ 2.90 GHz processor, 8 GB of RAM and 64-bit operating system. The computation load of algorithm is measured as follows [30]:

Fig. 8 3D performance of adaptive p Table 3 IAEs of $p = 1$ and adaptive p strategy

Strategy	IAE (m)
$p = 1$	1.653
Adaptive p	0.514

Table 4 GSs of $p = 1$ and adaptive p strategy

Strategy	GS (°)
$p = 1$	0.278
Adaptive p	0.105

$$T_c = \frac{T_{CPU}}{T_s},$$

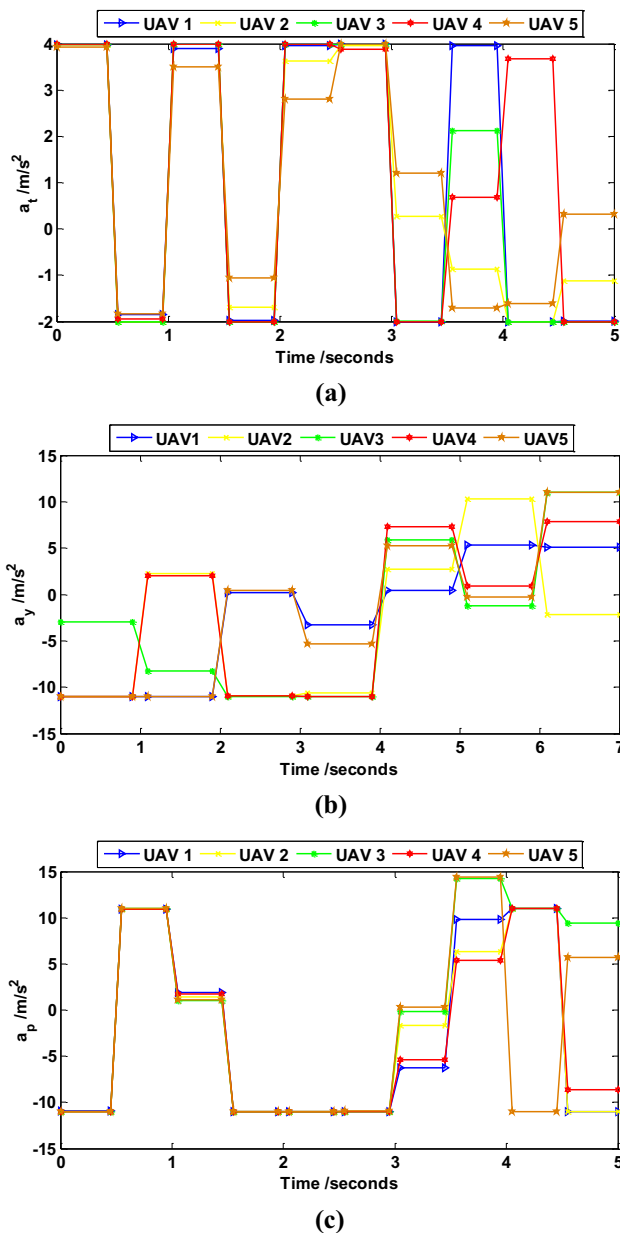


Fig. 9 Control variable for adaptive p : **a** a_t ; **b** a_y ; **c** a_p

Fig. 10 Adaptive selection of p

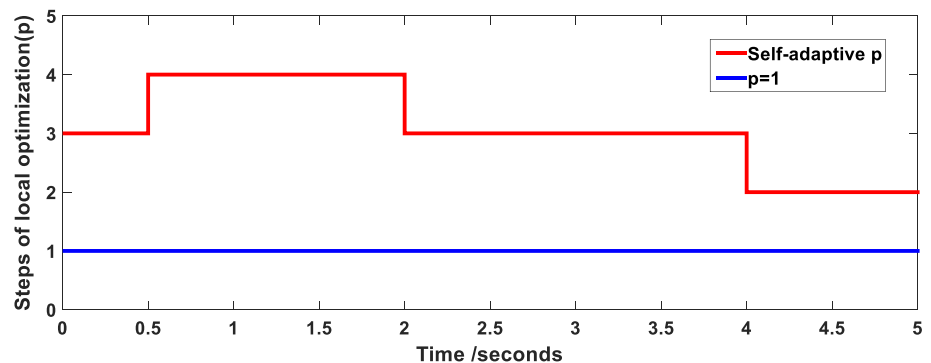


Table 5 Computation load of different algorithms

Algorithm	$T_c(s) (p = 1)$	$T_c(s) (\text{adaptive } p)$
Proposed algorithm	0.653	0.792
Grey wolf optimization algorithm	0.712	0.824
Particle swarm optimization algorithm	0.856	1.031

where T_c denotes the average computation time, T_{CPU} is the total measured CPU time, and T_s represents the total simulation time. For the implementable controller in real time, it is required that $T_c < 1$.

To better illustrate the computation load of proposed DE algorithm, we introduce the comparative simulations, where two other numerical algorithms are included, which denotes the grey wolf optimization algorithm [31] and particle swarm optimization algorithm [32]. The corresponding results are gathered in Table 5. From the table, it can be seen that for both $p = 1$ and adaptive p cases, the computation load of proposed DE algorithm is slightly less than that of two other algorithms.

6 Conclusion

In this paper, the distributed MPC with adaptive DE algorithm was proposed for the multi-UAV system achieving formation keeping and obstacle/collision avoidance effectively, as well as taking the state and input saturation into account. Different from other similar methods, the proposed control scheme combined the adaptive improvements in DE algorithm and introduced the adaptive selection for prediction horizon. Besides, the asymptotic convergence of the rolling optimization was analyzed. Finally, through simulation results, the effectiveness of the proposed control strategy was illustrated. However, the applicability of this paper is limited to the neglect of disturbances. Therefore, an appealing

extension of our work would focus on the possibilities of complex disturbances in the input or output channel.

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Compliance with Ethical Standards

Conflict of interest The authors declare that there are no conflicts of interest regarding the publication of this paper.

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